Basilisk MPI scaling for multiphase flows

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Abstract

Pressure projection couples the velocity field across an incompressible-flow calculation at every time step. Its multigrid Poisson solve therefore provides a natural test of CPU and MPI performance. We compare two stock Basilisk kernels and five multiphase-flow workloads on MareNostrum 5 and Snellius. The kernel measurements reach 16,384 MPI ranks; Marangoni-driven migration reaches 1,024 ranks and includes a terminal-velocity reference comparison. Uniform-grid bursting, sheet retraction and drop-impact calculations show how the useful rank count changes with resolution and constitutive model. The finest axisymmetric retraction and impact cases extend to 2,304 ranks, whereas the present three-dimensional viscoelastic-impact data cover one 192-core node. These fixed-size sweeps identify where additional ranks cease to reduce wall time. They also define the workloads and node-aligned measurements needed to extend the comparison to larger systems.

1 From pressure projection to multiphase flow

An incompressible Basilisk calculation advances advection, viscous stresses and interfacial forces, then projects the velocity onto a divergence-free field. This last step requires a multigrid Poisson solve. Restriction and prolongation transfer information between mesh levels, while MPI exchanges couple the subdomains. As the number of ranks increases at fixed problem size, each rank has less local work with which to offset these exchanges. The resulting wall-time curve locates the useful strong-scaling range.

We begin with the stock Poisson and Laplacian tests, retained without modification from Basilisk [1, 2]. We then add interface reconstruction, surface tension and, for the elastic cases, constitutive evolution. The Marangoni workload varies both mesh resolution and drop count, and is compared with a terminal-velocity reference. The other four application kernels measure scaling without a separate physical-validation study here. Source cases, compact timing tables and figure scripts are available at `https://github.com/comphy-lab/hpc-basilisk-scaling`.

2 Stock Poisson and Laplacian kernels

The two-dimensional test uses a full quadtree at $L = 14$, or 2^{28} cells. On Snellius, the Poisson time decreases through 8,192 ranks and plateaus at 16,384 ranks (Figure 1(a)). The growing MPI contribution relative to wall time shows that communication limits this part of the curve. The published Curie measurements provide a reference for the same calculation.

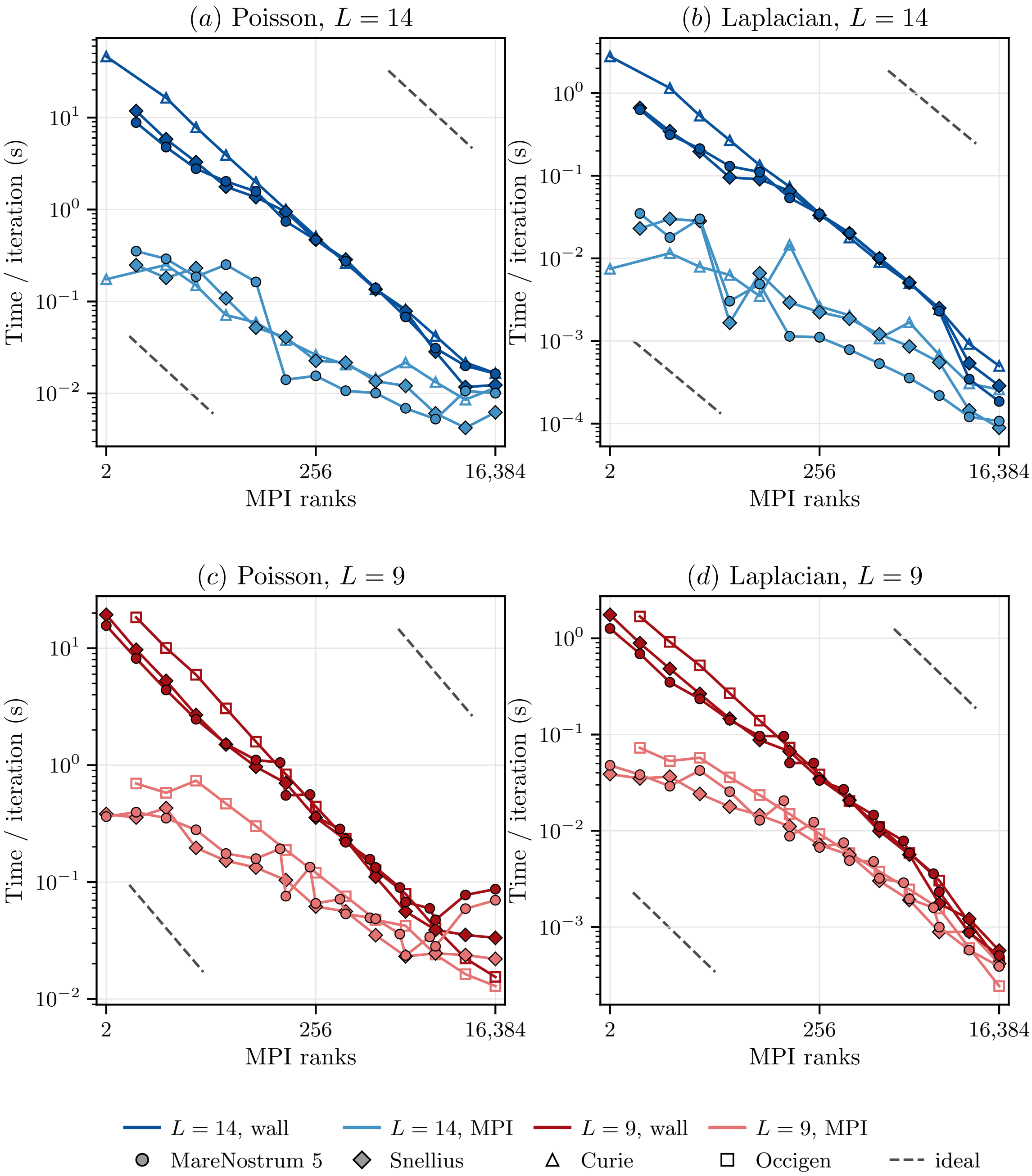

Figure 1: Strong scaling of the stock Poisson (left) and Laplacian (right) kernels. (a,b) Full two-dimensional quadtree at $L = 14$. (c,d) Full three-dimensional octree at $L = 9$. Blue and red identify $L = 14$ and $L = 9$, respectively; dark and light shades give wall and MPI time. Circles and diamonds identify MareNostrum 5 and Snellius. Open symbols show the Curie (two-dimensional) and Occigen (three-dimensional) reference data. Solid curves connect measurements. Paired grey dashed guides in the lower-left and upper-right corners show $T \propto N_{\mathrm{MPI}}^{-1}$ with arbitrary vertical normalization.

The three-dimensional test uses a full octree at $L = 9$, or 2^{27} cells. The Snellius Poisson time falls from 19.34 s at two ranks to 0.0390 s at 4,096 ranks, with only a further reduction to 0.0333 s at 16,384 ranks (Figure 1(c)). Occigen supplies the corresponding stock-test reference. Together, the wall and MPI timings distinguish the low-rank regime, governed primarily by local computation and memory access, from the large-rank regime in which communication takes a substantial fraction of the iteration.

3 Marangoni-driven migration

A surface-tension gradient drives the active drop. We use the conservative, balanced treatment of interfacial stresses described by Abu-Al-Saud, Popinet and Tchelepi [4]. The cluster case adds timing, restart and controlled output to the public Basilisk test while retaining its physical reference problem.

Increasing the resolution of an axisymmetric uniform mesh and increasing the number of drops in a planar domain provide two ways to enlarge the workload (Figure 2). In both cases, the larger problems continue to benefit from additional ranks after the smaller ones have saturated. These measurements stop at 1,024 ranks, equivalent to about 5.3 fully populated 192-core nodes.

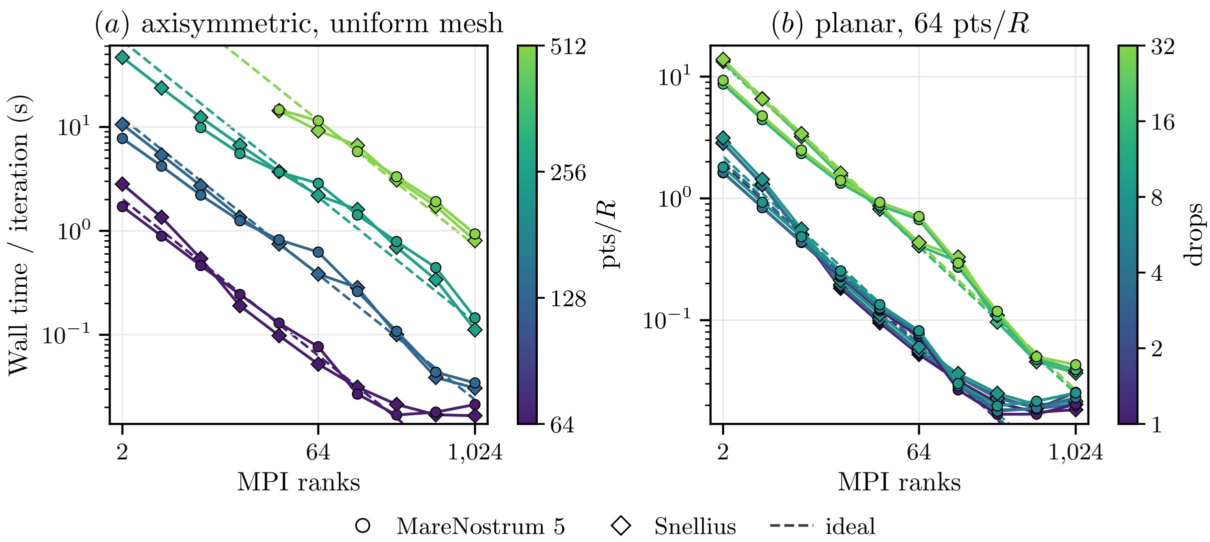

Figure 2: Active-drop scaling in the abundant-fuel limit on MareNostrum 5 GPP (circles) and Snellius Genoa (diamonds). (a) Axisymmetric uniform meshes at 64–512 points per radius. (b) Planar workloads with 1–32 drops at 64 points per radius. Dashed lines show ideal scaling. Integral surface-tension formulations for Marangoni flows are described in Ref. [5].

Figure 3 compares the computed migration speed with the Young–Goldstein–Block prediction and the Basilisk reference data. The relative error decreases at approximately second order through 32 points per radius. Finer meshes bring the speed closer to the analytical value, but the error plateaus; a single second-order fit across all resolutions would conceal this limit. The velocity fields show the development of the flow in the drop frame.

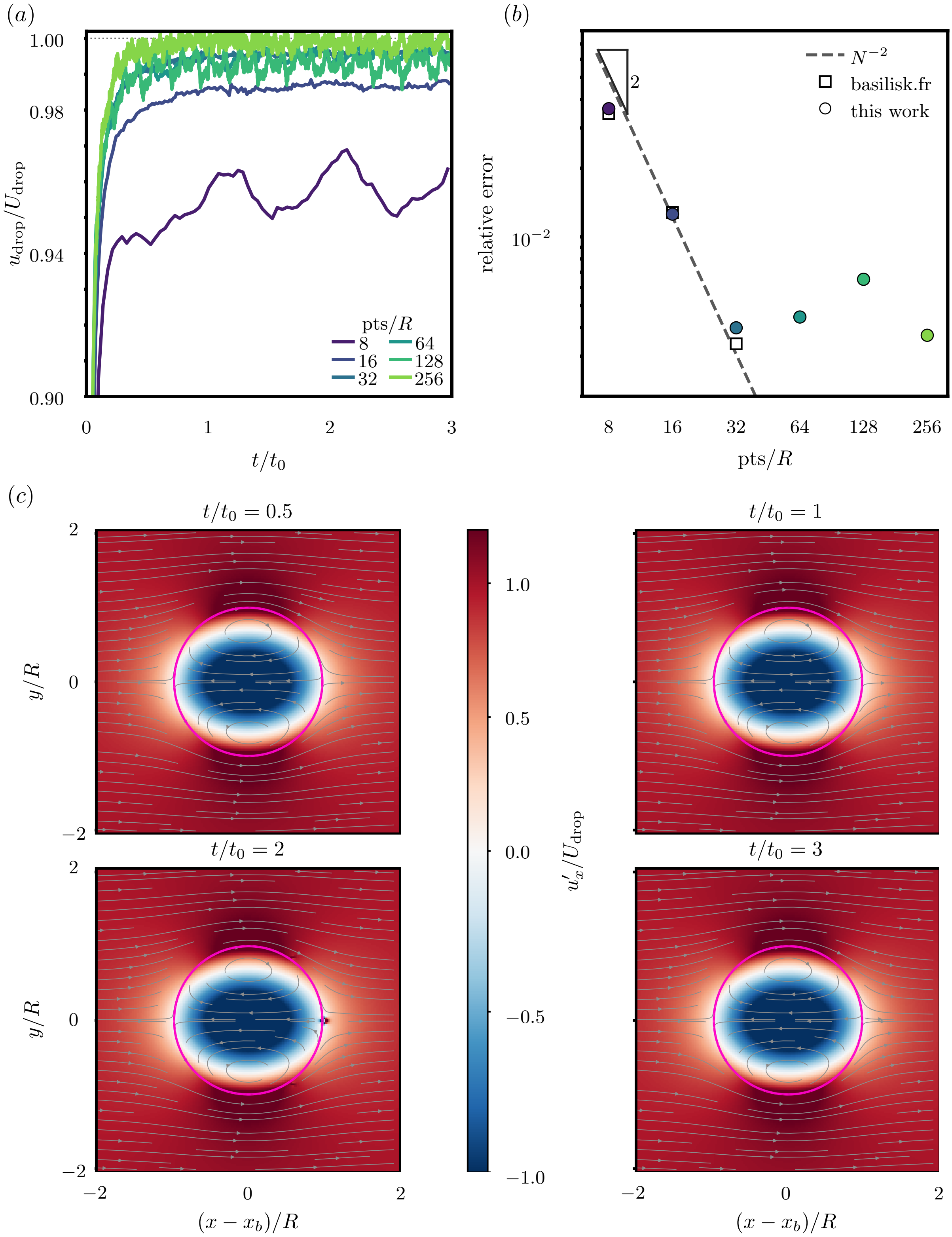

Figure 3: Active-drop migration in the abundant-fuel limit. (a) Speed relative to the Young–Goldstein–Block prediction at 8–256 points per radius. (b) Relative error and the coarse-grid N^{-2} trend. (c) Drop-frame axial velocity at 64 points per radius, shown at $t/t_0 = 0.5$, 1, 2 and 3. Magenta marks the interface. See Ref. [5] for the interfacial-stress formulation.

4 Uniform-grid multiphase-flow kernels

The remaining kernels retain the uniform-tree MPI decomposition and change the flow: bubble bursting, elastic Taylor–Culick retraction, Newtonian drop impact and three-dimensional viscoelastic impact. Each timing covers ten incompressible two-phase solver steps. Geometric initialization is excluded, as are the serial interface reconstruction and seed dump for bursting. The MPI binaries write no checkpoints during these runs.

The three axisymmetric cases use $L = 9$, 10 and 11, corresponding to 512^2, 1024^2 and 2048^2 cells. The $L = 9$ and $L = 10$ sweeps cover 2–768 ranks; the $L = 11$ sweeps start at eight ranks. Retraction and impact extend to 2,304 ranks at $L = 11$, while bursting already turns upward at 768 ranks. The three-dimensional impact uses a full $L = 7$ octree (128^3 cells) over 2–192 ranks.

At $L = 10$, the three axisymmetric kernels have similar scaling ranges, despite their different per-step costs (Table 1). At $L = 9$, wall time approaches a floor near 0.008 s for bursting beyond 96 ranks, 0.010 s for retraction and 0.011 s for impact. Refinement extends the useful rank range; the finest retraction series still decreases at its 2,304-rank endpoint.

	Wall time per iteration (s), $L = 10$		
Kernel	2 ranks	192 ranks	768 ranks
Bursting bubble	1.28	0.0174	0.0108
Elastic Taylor–Culick	1.85	0.0231	0.0127
Newtonian drop impact	2.04	0.0267	0.0173

Table 1: Selected Snellius Genoa timings from the fixed-size axisymmetric sweeps.

Bursting at $L = 11$ follows the ideal trend through 384 ranks (0.0407 s), then rises to 0.0781 s at 768 ranks (Figure 4). The elastic sheet adds log-conformation extra-stress evolution to the projection step. Its $L = 11$ time decreases from 1.90 s at eight ranks to 0.0341 s at 768 ranks and 0.0193 s at 1,536 ranks, with a smaller further reduction to 0.0152 s at 2,304 ranks (Figure 5).

The Newtonian impact places the drop at $(1.05, 0)$ in a box of side 8, with velocity towards the left wall and the axis of symmetry on the bottom boundary. At $L = 11$, its time follows the ideal trend to 0.0339 s at 768 ranks and 0.0250 s at 1,536 ranks, then rises to 0.0423 s at 2,304 ranks (Figure 6). Both finest-resolution sweeps depart from ideal scaling. Retraction continues to become faster, whereas impact turns upward within the measured range.

For three-dimensional viscoelastic impact, the $L = 7$ time falls from 10.8 s at two ranks to 0.331 s at 192 ranks, a factor of 32 (Figure 7). This curve remains monotonic but departs from ideal scaling earlier than the axisymmetric cases. The application timings do not separate computation from communication and other overheads, so they cannot identify the cause of this departure.

The temporal sequences in Figures 4–7 come from completed simulations chosen to show the flow geometry and fields. They are independent of the ten-step timing trajectories. Each sequence reads left to right, then top to bottom, with a common spatial scale. Each scalar field uses one colour normalization across all four states. The three-dimensional sequence shows coalescence-induced jumping; its timing panel measures viscoelastic impact.

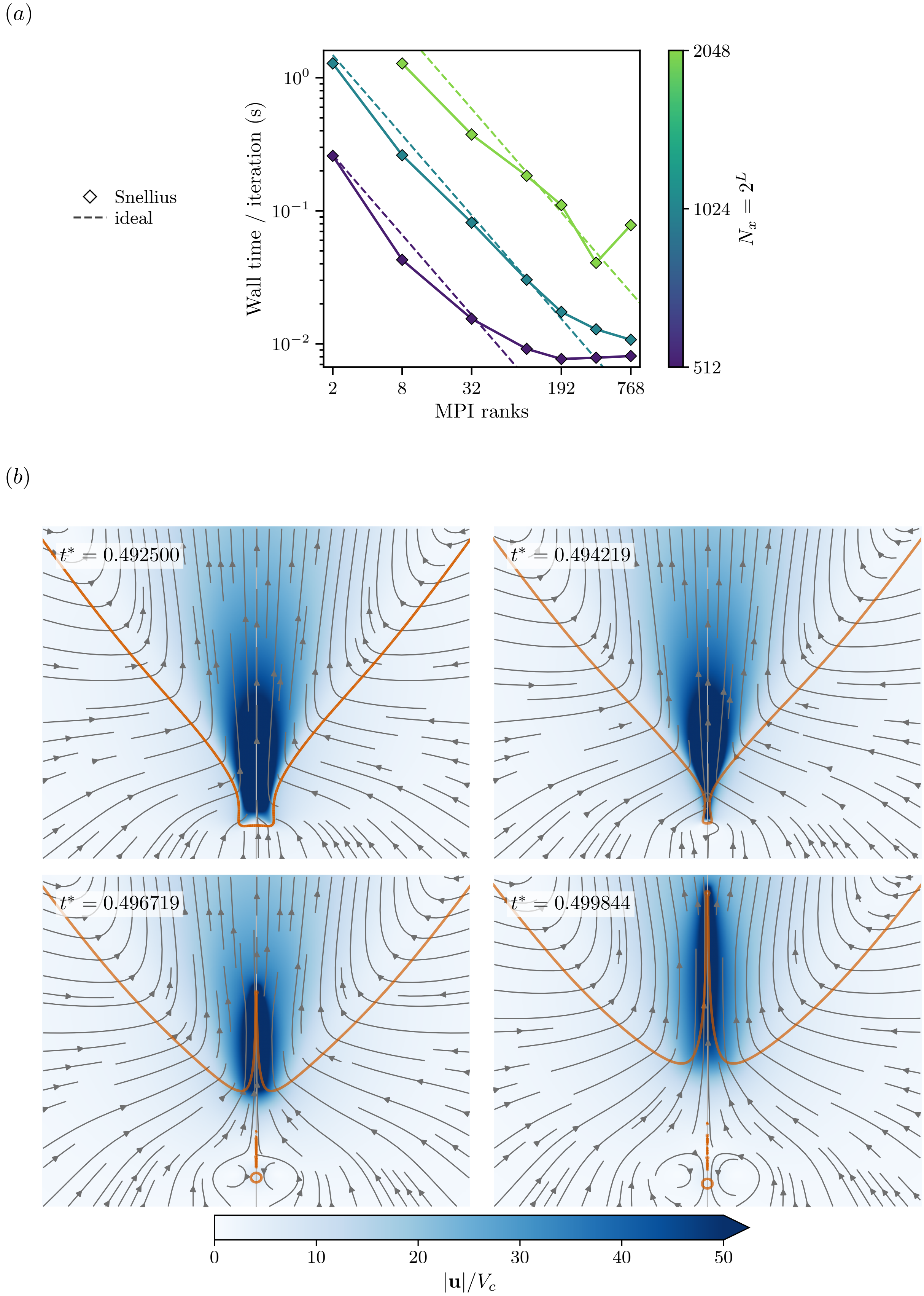

Figure 4: Bursting bubble [6, 7]. (a) Snellius Genoa strong scaling of the axisymmetric uniform-grid kernel; colour gives $N_x = 2^L$ and dashed lines show ideal scaling. (b) Jet inception [8] at $Oh = 0.03$, $Bo = 0$. Blue gives $|\boldsymbol{u}|/V_c$, where $V_c = \sqrt{\sigma/(\rho_l R)}$ and $t^* = tV_c/R$; grey curves are instantaneous streamlines and orange marks the interface. The sequence shows the narrowing cavity, emerging jet and entrapped bubble. Colours saturate above $50V_c$.

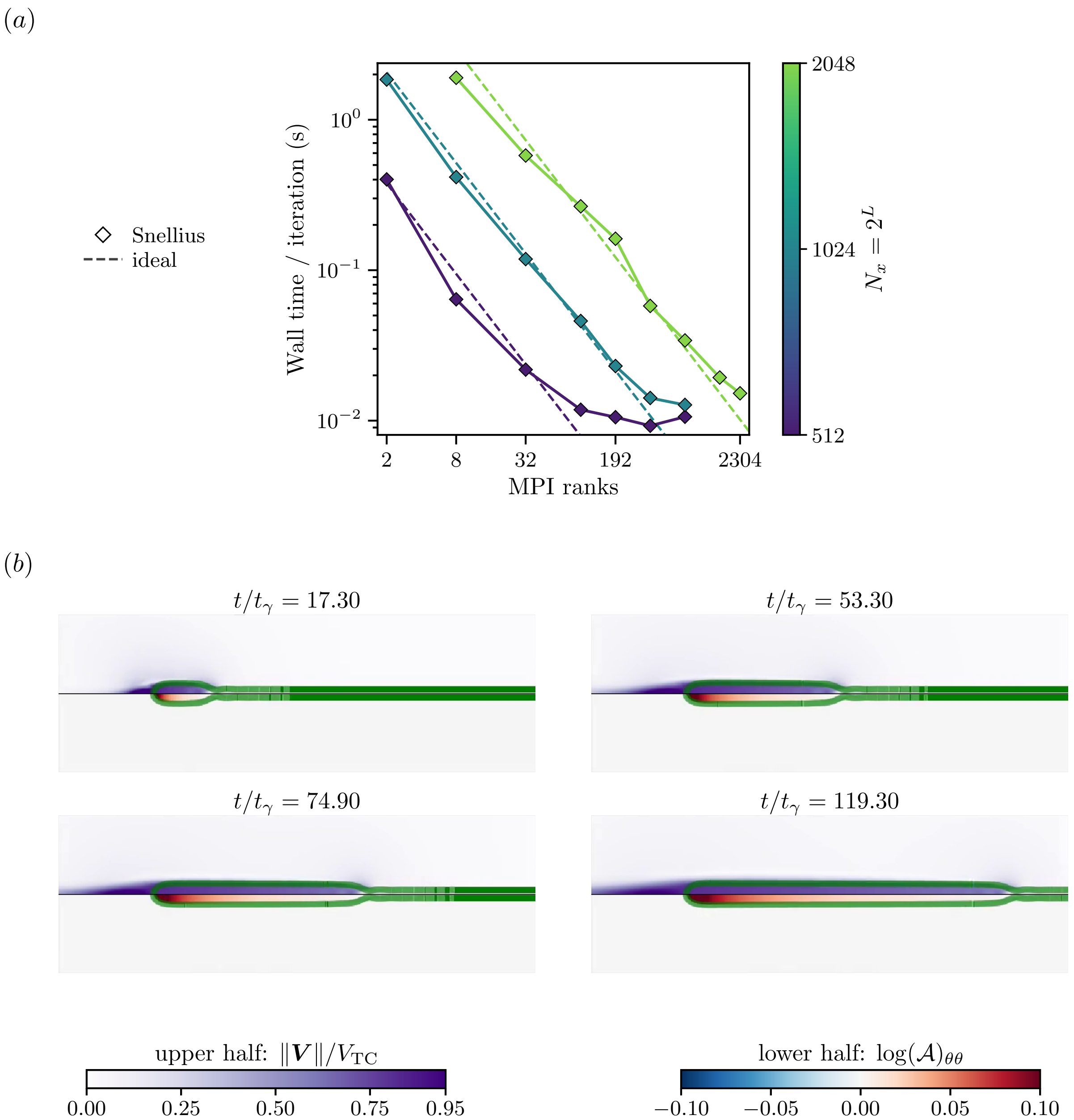

Figure 5: Elastic Taylor–Culick retraction. (a) Snellius Genoa strong scaling of the axisymmetric uniform-grid kernel. (b) Rim evolution at $Ec = 0.1$, $Oh = 0.05$ in a relaxation-free calculation. The view follows the rim. Each frame combines speed relative to the Taylor–Culick velocity (upper half) and the azimuthal log-conformation $(\log \mathcal{A})_{\theta\theta}$ (lower half); green marks the interface. Here $t_\gamma = \sqrt{\rho H_0^3/\sigma}$, with H_0 the initial sheet thickness. For retraction and the influence of the surroundings, see Ref. [9].

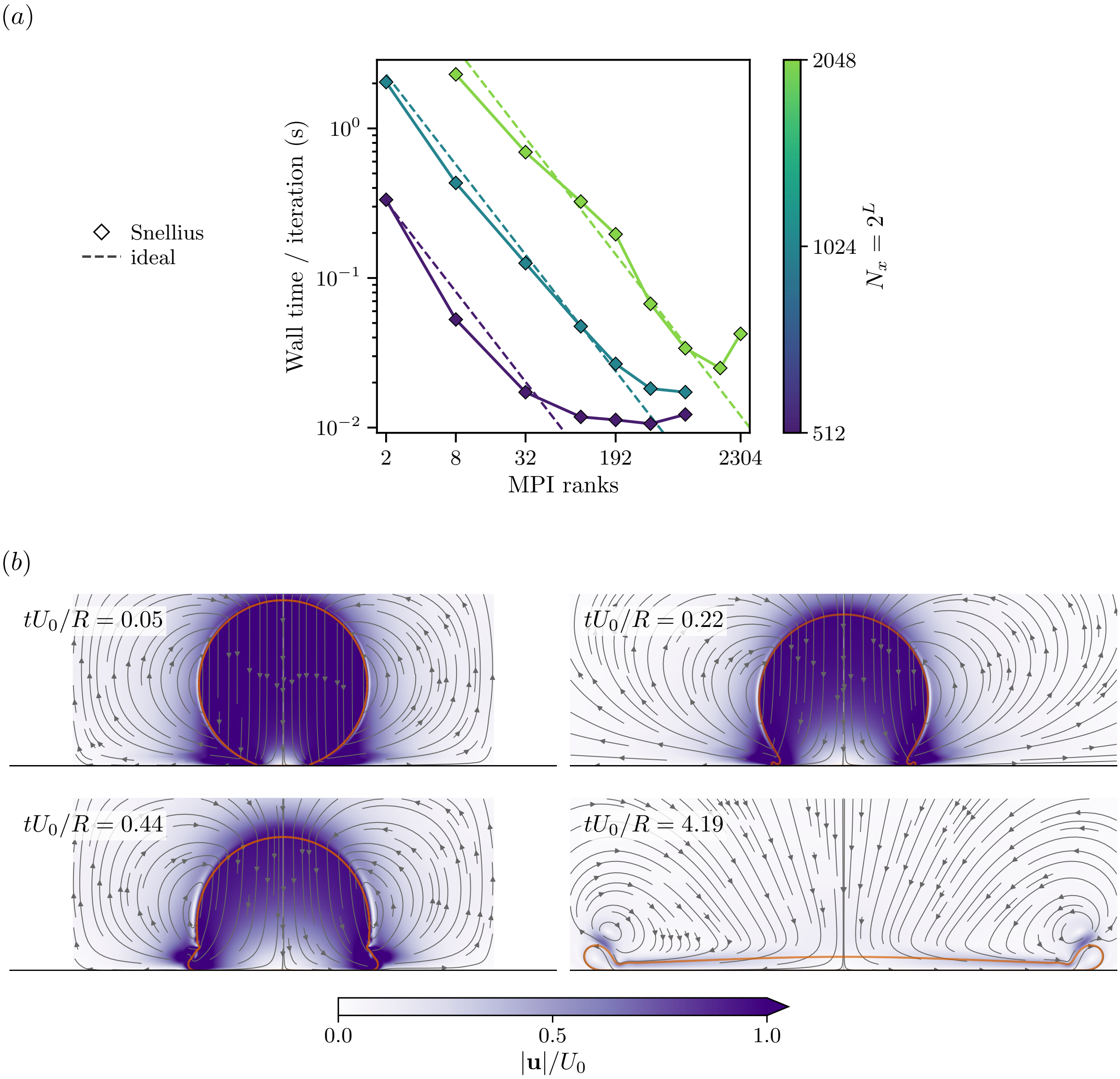

Figure 6: Newtonian drop impact. (a) Snellius Genoa strong scaling of the axisymmetric uniform-grid kernel. (b) Impact and spreading at diameter-based $We_D = 200$ and $Oh_D = 0.01$. Lengths are scaled by the initial drop radius R and time by R/U_0, where U_0 is the impact speed. Purple shows speed, grey curves are instantaneous streamlines, orange marks the interface and black marks the substrate. Colours saturate above U_0. All four frames retain the same view, including the extended lamella at $tU_0/R = 4.19$. Related studies address free-surface wave focusing [10] and drop-impact forces and spreading [11, 12].

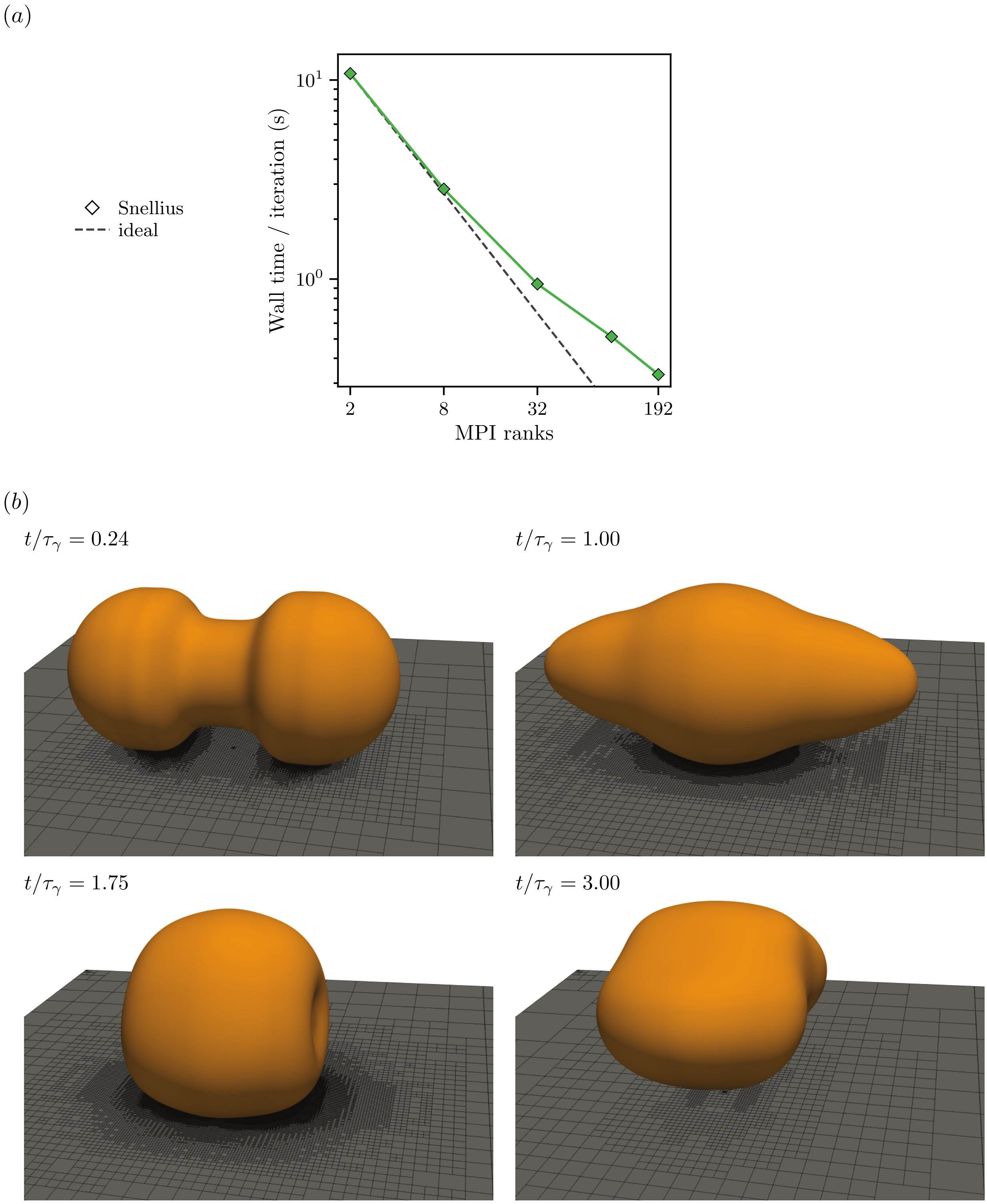

Figure 7: Three-dimensional calculations. (a) Snellius Genoa strong scaling of viscoelastic drop impact on a full $L = 7$ octree. (b) Coalescence-induced jumping of two equal drops at $Oh = 0.05$ and $Bo = 0$, from a separate calculation. Orange shows the free surface; the substrate plane shows the adaptive mesh. Times are scaled by $\tau_\gamma = \sqrt{\rho R^3/\sigma}$, where R is the initial radius of either drop. The coalescence sequence illustrates three-dimensional interface motion and is not an output of the impact benchmark in (a).

5 Extending the comparison to larger systems

The kernel sweeps use powers of two in rank count. Their upper limit of 16,384 ranks corresponds to 85.3 fully populated 192-core nodes: it brackets 64 nodes, but does not reach the class above 100 nodes. A 12,288-rank run is needed to measure 64 fully populated nodes directly. Table 2 sets out the corresponding node-aligned tests.

Scale	Nodes	MPI ranks	Coverage in the present data
Small	1, 2, 4, 8, 10	192–1,920	Kernel curves bracket the range; Marangoni reaches 1,024 ranks; axisymmetric retraction and impact reach 2,304
Medium	16, 32, 64	3,072–12,288	Kernel measurements at 4,096 and 8,192 ranks bracket 64 nodes together with the 16,384-rank endpoint
Large	128, 256	24,576–49,152	Unmeasured; problem size depends on memory use and the useful scaling range

Table 2: Node-aligned benchmark points for 192 CPU cores per node. The node counts are proposed measurements; the last column describes the existing rank sweeps.

The existing $L = 14$ quadtree and $L = 9$ octree kernels can be retained at small scales. For 16–64 nodes, an $L = 10$ viscoelastic-impact calculation is the next three-dimensional candidate, beginning at 16, 32 and 64 nodes subject to a memory check. If that problem leaves too little work per rank, an $L = 11$ octree can be considered at 128 nodes. A short pilot must establish memory use and initialization cost before this increase; extending beyond 128 nodes remains optional. The problem must retain enough local work for its rank sweep to resolve a scaling regime.

Application comparisons also require the same domain size, interface area and output policy on every machine. The two-phase case should accompany the stock kernels at small and medium scales; these workload choices must be fixed before extending the application to the large-node range.

6 Reproducing and interpreting the measurements

A timing is tied to its source revision, compiler and optimization flags, MPI implementation and launcher. Its hardware description must include the node type, ranks per node, thread affinity and rank binding. Mesh dimension, level, cell count and cells per rank define the numerical workload; wall time, MPI time, iteration count and excluded initialization define what was timed. These quantities allow the same calculation to be compared across machines.

The committed timing tables contain one observation per configuration. They locate scaling regimes but do not quantify run-to-run variability. Compact image inputs, source parameters, attribution and scripts accompany the figures. The illustrated flow evolution provides physical context; performance conclusions follow from the timing data.

7 Conclusion

The stock kernels resolve the transition from local computation to a communication-limited regime. Marangoni migration adds a reference comparison and shows how resolution and drop count shift the useful rank range. The uniform-grid cases extend the comparison to jetting, elastic retraction and impact: at fixed resolution, additional ranks eventually yield little benefit and can increase wall time. The present data therefore support small-scale application comparisons and bracket the 64-node range through the stock kernels. Larger three-dimensional workloads and exact node-aligned measurements are needed to extend those conclusions above 100 nodes.

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